

homework-2

#packages

Question 1

```
install.packages("aRtsy")
```

`install.packages()` is used specifically to install a specific package onto your computer and only needs to be run when installing the package for the first library. The `library()` function's purpose is to load an already existing package into the current session.

Question 2

`library(aRtsy)`

The Collatz conjecture is a famous unsolved math problem that states that if you repeatedly apply the rules of if a number is even: divide it by 2, and if it is odd: you would multiply it by 3 and add 1. If you repeatedly apply these rules you will always get 1.

Collatz conjecture Latex equation:

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases}$$

Question 3:

This function creates a visual representation of the Collatz conjecture. The “n” argument determines the number of starting values, while the color shows what color

part B:

- You get a different output every time because the function uses randomization. Although it is the same code, there are different random values generated, leading to different results
- It would be troublesome to put this code into a .qmd file because the random values of function can be different, leading to varying results, which means the output of the function will look different each time it is rendered
- The `set.seed()` function essentially lets us control R's number generation. By setting a seed, R will consistently produce the same random values each time the code is run.

```
set.seed(13)

canvas_collatz(
  colors = c("blue"),
  n = 10
)
```



Question 4:

```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```



- The code in the previous question creates a visualization of the Collatz conjecture based on one starting value. The code in question 4, provides us numbers 1-20, so the function creates a visualization using 20 different starting values. The result would be a more detailed image as more sequences are being used.
- It would make sense to use `set.seed(1)`, as we would want to be able to produce the same image consistently when it comes to rendering. Through using this function we will be able to control the random number generation and be able to control the randomness.